Saad Murtaza Khan

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researchgate

October 28, 2026

Summary

I am a Bioinformatics Scientist with over 10 years of experience in next-generation sequencing, including six years focused on single-cell and spatial transcriptomics in tumor biology. I currently work as a Bioinformatics Specialist at Massachusetts General Hospital, where I analyze single-cell and spatial data to define tumor cell states and how they are organized in relation to the tumor microenvironment in brain tumors. My goal is to turn computational insights into a better understanding of how tumors develop and progress, and to help guide the development of immunotherapies.

Skills

- **Machine Learning & Agentic AI**: PyTorch, TensorFlow, scikit-learn; convolutional networks for sequence and structure; LLM application development (LangGraph, Streamlit, Anthropic API), limma
- **Single-Cell & Spatial Analysis**: Cell Ranger, Space Ranger, Xenium Ranger, Seurat, Scanpy, SpatialExperiment, BANKSY, RCTD, CellTrek, Cell2location, CellChat, nichenetr, Monocle, CellRank, SingleR, InferCNV, CITE-seq
- **Somatic variant calling and annotation**: BWA, GATK, VarScan2, Strelka, ANNOVAR, VEP;
- **HPC, Workflows & Software**: SLURM, XSEDE/TACC, WDL, Cromwell, Terra, Pegasus; Docker, Singularity; git
- **Programming**: Python, R, C, Perl
- **Visualization**: ggplot2, Plotly, ComplexHeatmap, seaborn, Shiny, Streamlit, Cytoscape
- **Databases**: TCGA, GTEx, GEO, cellxgene
- **Languages**: English, Hindi, Urdu

Research Experience

- **Brain Tumor Research Center, Massachusetts General Hospital** — Boston, MA
  Bioinformatics Specialist — *11/2022 – present*
  - **Spatial architecture of the glioblastoma immune microenvironment**: Led the analysis of a 22-sample Visium cohort with matched Xenium and single-nucleus RNA-seq, mapping tumor and immune cell states from tumor core to infiltrating edge. Used BANKSY for niche detection, RCTD and CellTrek for deconvolution, and CellChat and NicheNet for niche-restricted ligand-receptor inference. Identified interactions confined to specific anatomical niches, which argues that effective combinations will need to be chosen for the region they are meant to act on. First author on the resulting preprint.
  - **Single-cell and exome characterization of meningioma**: Combined single-cell RNA-seq and whole-exome sequencing to link driver mutations to tumor and immune cell gene expression programs at single-cell resolution.
  - **Epitranscriptomics of glioma**: Analyzed long-read nanopore RNA-seq to define glioma subtype-specific m6A landscapes and their effect on isoform usage and gene expression, separating the contributions of individual m6A regulators.
- **Brain Tumor Center, Washington University St. Louis** — St. Louis, MO
  Bioinformatics Scientist — *12/2019 – 11/2022*
  - **Determinants of response to PD-1 blockade in brain tumors**: Analyzed single-cell transcriptomes from syngeneic glioma models to explain divergent checkpoint blockade response. GL261 tumors were enriched for PD-1-responsive progenitor exhausted $CD8^+$ T cells, while CT2A tumors were dominated by PD-1-refractory terminally exhausted $CD8^+$ T cells, and CD4 T cell depletion shifted the exhaustion trajectory. Co-first author in the Journal for ImmunoTherapy of Cancer.
  - **Myeloid checkpoint target characterization**: Contributed the single-cell analysis supporting TREM2 as a myeloid checkpoint in glioblastoma, showing how TREM2 inhibition shifts myeloid cells toward an antitumor state (Science Advances).

- **Genetic and transcriptional evolution at AML relapse**: Integrated single-cell RNA-seq with enhanced whole-genome sequencing across paired presentation and relapse samples to link subclonal genetic change to transcriptional state. Identified relapse-enriched cell states and quiescent, stem-like subpopulations that are candidate sources of resistance.
- **Single-cell atlas of the vestibular schwannoma ecosystem**: Built a single-cell atlas of schwannoma and its microenvironment and identified a nerve injury-like program associated with myeloid infiltration, monocyte recruitment, and larger tumor size, nominating a mechanism and candidate targets (Nature Communications).
- **Reproducible pipeline infrastructure for single-cell analysis**: Designed and implemented modular WDL pipelines covering preprocessing, quality control, normalization, clustering, and differential expression, deployed on cloud and HPC through Cromwell and Terra and published as a public Dockstore collection used by the lab and outside groups. dockstore.org/organizations/PettilabMGH

- **Informatics Institute, University of Missouri** — Columbia, MO
  Graduate Research Assistant — *09/2011 – 12/2019*
  - **Deep learning for RNA modification prediction**: Developed MU-PseUDeep, a convolutional neural network integrating primary sequence with predicted secondary structure to identify pseudouridine sites, outperforming existing methods across human, mouse, and yeast. First method to incorporate RNA secondary structure for this task. Genome-wide screening showed enrichment of predicted sites in neurodegeneration-associated transcripts.
  - **Large-scale variant analysis for 1000 genome Soybean project**: Built a parallelized variant calling and annotation workflow on the Pegasus workflow management system, running GATK across soybean genomes on XSEDE high-performance computing at the Texas Advanced Computing Center, and delivered the results as a queryable resource in SoyKB.
  - **Cancer genomics and microRNA profiling in JMML**: Analyzed whole-exome and high-depth panel sequencing from a JMML patient, identifying PI3K/AKT pathway mutations alongside known RAS/MAPK alterations, and paired this with microRNA profiling showing upregulated oncomiRs targeting PI3K/AKT genes.
  - **Soybean epigenomics**: Analyzed whole-genome bisulfite and RNA-seq data to characterize inheritance of MAMP-triggered immunity in soybean, identifying heritable differentially methylated regions including variants with non-Mendelian inheritance. First genome-wide methylation survey in soybean.

- **Thermo Fisher Scientific Inc.** — South San Francisco, CA
  Bioinformatics Algorithm Validation Engineer Intern — *06/2017 – 08/2017*
  - **Performance benchmarking and automated regression testing for Ion Reporter**: Compared workflow performance across Ion Reporter 5.2, 5.4, and 5.6, wrote Perl log parsers to profile per-actor runtime and R scripts to visualize the comparisons, and built automated regression suites in the in-house framework for functional annotation, BRCA CNV calling, and Oncomine fusion (OCAv3) workflows.

Education

- **University of Missouri - Informatics Institute** — Columbia, MO
  PhD Informatics — *08/2011 - 12/2019*
  (Advisor: Prof. Dong Xu, (GPA: 3.8)

  Dissertation: Informatics Approaches in Epigenomics and Epitranscriptomics

- **Jamia Millia Islamia - Department of Computer Science** — New Delhi, India
  Master of Science Degree in Bioinformatics — *08/2006 - 07/2008*
  (Advisor: Dr. Punit Kaur) (GPA: 4)

  Thesis: Clinically oriented database of Diabetes Mellitus

- **Aligarh Muslim University - Faculty of Life Sciences** — Aligarh, India
  Bachelor of Science (Honors) Degree in Zoology; (GPA: 3.8) — *08/2002 - 07/2005*

Publications & Preprints

- **Saad M. Khan***, Anthony Z. Wang*, Rupen R. Desai*, et al. Mapping the spatial architecture of glioblastoma from core to edge delineates niche-specific tumor cell states and intercellular interactions. bioRxiv 2025.04.04.647096 (2025) doi:10.1101/2025.04.04.647096. (*Equal contribution)
- Clara Alves-Pereira, Gyeong Dae Kim, Ngima Sherpa, Kayla Colvin, **Saad M. Khan**, Khanh P. Phan, Anthony Z. Wang, et al. Integrated Spatial and Single-Cell Transcriptomic Analysis of Aggressive Glioblastoma Growth Dynamics. bioRxiv 2026.05.11.724432 (2026) doi:10.64898/2026.05.11.724432.
- Syeda Maheen Batool, Hanna Lee, Koushik Muralidharan, **Saad Murtaza Khan**, Ana K. Escobedo, Denalda Gashi, Kesli Faber, et al. IDH1-Associated m6A Methylation Is Linked to Transcriptomic Heterogeneity in Glioma. Cancers 18(11):1825 (2026). doi:10.3390/cancers18111825

- Jose A. Maldonado, Bryce L. Mashimo, Anthony Z. Wang, Rupen Desai, **Saad M. Khan**, et al. Single-Cell Analysis of Meningiomas Reveals Mutation-Associated Tumor and Immune Cell Gene Expression Programs. bioRxiv 2025.05.21.655421 (2025) doi:10.1101/2025.05.21.655421.
- Anthony Z. Wang, Bryce L. Mashimo, Maximilian O. Schaettler, Ngima D. Sherpa, Lydia A. Leavitt, Alexandra J. Livingstone, **Saad M. Khan**, et al. (2024) Glioblastoma-infiltrating CD8+ T cells are predominantly a clonally expanded GZMK+ effector population. Cancer Discov. https://doi.org/10.1158/2159-8290.CD-23-0913
- Thomas F. Barrett, Bhuvic Patel, **Saad M. Khan**, et al. (2024) Single-cell multi-omic analysis of the vestibular schwannoma ecosystem uncovers a nerve injury-like state. Nat Commun 15:478. https://doi.org/10.1038/s41467-023-42762-w
- Rui Sun, Rowland Han, Colin McCornack, **Saad M. Khan**, et al. TREM2 inhibition triggers antitumor cell activity of myeloid cells in glioblastoma. (2024) Sci Adv 9:eade3559. https://doi.org/10.1126/sciadv.ade3559
- **Saad M. Khan***, Rupen Desai*, et al. Impact of CD4 T cells on intratumoral CD8 T-cell exhaustion and responsiveness to PD-1 blockade therapy in mouse brain tumors. (2022) J Immunother Cancer 10:e005293. https://doi.org/10.1136/jitc-2022-005293 (*Equal contribution)
- Allegra A. Petti, **Saad M. Khan**, et al. Genetic and Transcriptional Contributions to Relapse in Normal Karyotype Acute Myeloid Leukemia. (2022) https://doi.org/10.1158/2643-3230.BCD-21-0050
- **Saad M. Khan**, Fei He, et al. MU-PseUDeep: A deep learning method for prediction of pseudouridine sites. (2020) Comput Struct Biotechnol J 18:1877–1883. https://doi.org/10.1016/j.csbj.2020.07.010
- **Saad M. Khan**, Jason E Denney, et al. Whole-exome sequencing and microRNA profiling reveal PI3K/AKT pathway's involvement in juvenile myelomonocytic leukemia. (2018) Quant Biol 6:85–97. https://doi.org/10.1007/s40484-017-0125-2
- Yang Liu*, **Saad M. Khan***, Juexin Wang*, et al. PGen: Large-scale genomic variations analysis workflow and browser in SoyKB. (2016) BMC Bioinformatics 17:337. https://doi.org/10.1186/s12859-016-1227-y (*Equal contribution)
- Robert J Schmitz, Yupeng He, Oswaldo Valdés-López, **Saad M. Khan**, et al. Epigenome-wide inheritance of cytosine methylation variants in a recombinant inbred population. (2013) Genome Res 23:1663–1674. https://doi.org/10.1101/gr.152538.112
- Please visit my Google Scholar profile for the complete list of publications.

Software and Data Resources

- **Petti Lab single-cell WDL collection**, Dockstore: modular, containerized WDL workflows for scRNA-seq preprocessing through differential expression, portable across Terra, cloud, and HPC. dockstore.org/organizations/PettilabMGH
- **MU-PseUDeep**, convolutional neural network for pseudouridine site prediction from sequence and predicted secondary structure, with trained models and inference code. https://github.com/smk5g5/MU-PseUDeep
- **PGen**, parallelized variant analysis workflow for large genome collections on XSEDE HPC. https://github.com/pegasus-isi/PGen-GenomicVariations-Workflow
- Analysis code and containers: github.com/smk5g5 and hub.docker.com/u/smk5g5

Review Activities

- **Journal reviews**: reviewed papers for Cancer Cell International, Disease Markers and Frontiers in Genetics.

Talks and Poster Presentations

- Brain Tumor Research Center Floor Talks, Massachusetts General Hospital (2025)
- Center for Genomic Medicine (CGM) Retreat, Massachusetts General Hospital (2025)
- ISMB (2016), IEEE BIBM (2017), MCBIOS Symposium (2013), Missouri Life Sciences Week (2015–2017)
- MU Informatics Symposium, University of Missouri (recurring presenter, 2012–2018)

Service and Leadership

- Preparing Tomorrow's Leaders for Science (PTLS) (2016–2017)
- Organizing Committee, MU Informatics Symposium (2012–2014); Volunteer (2015–2016)
- Treasurer, Informatics Institute Graduate Student Association (MUII) (2012–2014)

References

- **Dr. Allegra Petti** Email: apetti@mgh.harvard.edu
  Assistant Professor, Department of Neurosurgery, Massachusetts General Hospital *(617) 306-3143*
- **Dr. Dong Xu** Email: dxu1@usf.edu
  Professor, Health Informatics Institute, University of South Florida *(573) 808-2219*
- **Dr. Albert H. Kim** Email: alberthkim@wustl.edu
  Director, Brain Tumor Center, Washington University, St. Louis *(314) 747-6561*
- **Dr. Jeffrey N. Bryan** Email: jnb33619@uga.edu
  Professor & Dept. Head, Small Animal Medicine & Surgery, UGA College of Veterinary Medicine